CRediT Contributor Roles Parser & Normalizer

Comprehensive Quality Assurance & Edge-Case Validation Scenario Specification

Objective: Live taxonomy harvesting, docx parsing, author resolution & CRediT mapping

Live Taxonomy Source: `https://credit.niso.org/` (Strictly runtime fetched)

Input Formats: Word Manuscripts (`.docx`) with variable sections

Output Format: Normalized Tabular Contributorship Dataset (`.csv`)

1. Overview & Core Operational Constraints

This document defines the comprehensive scenario list for testing an automated system designed to ingest academic manuscript Word files, dynamically harvest the official 14 CRediT (Contributor Roles Taxonomy) definitions directly from `credit.niso.org` at runtime, resolve author identities against document bylines, normalize diverse syntactic and narrative phrasing into standard CRediT roles, and output an audit-ready CSV file.

Mandatory Constraint — Zero Hardcoding: The 14 official CRediT roles must not be statically baked into regular expressions, dictionaries, or constants in code. They must be scraped dynamically from `https://credit.niso.org/` at runtime on each run or cached strictly with live timestamp verification.

2. Detailed Test Scenarios by Architectural Component

Component A: Dynamic Taxonomy Harvesting (`credit.niso.org`)

Scenario A.1: Canonical 14-Role Dynamic Ingestion

LIVE SCRAPE

Intent: Connect to `https://credit.niso.org/` at runtime and parse the 14 official roles without hardcoding.

Validation: The harvester extracts exactly 14 unique roles: *Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft,* and *Writing – review & editing*.

Scenario A.2: HTML Entity & Unicode Dash Normalization

EDGE CASE

Intent: Ensure scraper decodes HTML entities and normalizes punctuation.

Validation: `&` is parsed cleanly as `&` (e.g., *"Writing – review & editing"* → *"Writing – review & editing"*), and Unicode en-dashes (– , `–`), em-dashes (—), and standard hyphens (-) are reconciled to standard canonical representations.

Scenario A.3: Definition & Scope Harvesting for Semantic Matching

STANDARD

Intent: Scrape the detailed descriptions under each role link (e.g., `/contributor-roles/conceptualization/`) to feed the semantic similarity model rather than matching on role titles alone.

Validation: Definitions such as *"Ideas; formulation or evolution of overarching research goals and aims"* are extracted to enable matching phrases like *"designed the study"* to *Conceptualization*.

Scenario A.4: Network Outage & Throttling Resilience

RESILIENCE

Intent: Handle network unavailability, HTTP 429/503 errors, or server timeouts gracefully.

Validation: The system logs informative network failure logs, employs retry logic with backoff, and falls back to a locally cached live snapshot if previously fetched, accompanied by an explicit audit warning.

Component B: Manuscript File Ingestion & Integrity Checks

Scenario B.1: Standard XML Body Paragraph Extraction

STANDARD

Intent: Open valid Word OpenXML files (`.docx`) and extract ordered body paragraphs, headings, and runs preserving text sequence.

Validation: All paragraphs in files like `merged_01.docx` through `merged_11.docx` are parsed with zero lost characters.

Scenario B.2: Corrupted & Non-Zip Document Handling

NEGATIVE TEST

Intent: Validate behavior when encountering an invalid or damaged file (e.g., `merged_12.docx` , an arbitrary binary non-zip file).

Validation: Pipeline does not terminate with an unhandled exception. Instead, it catches the file error, logs `Error: Invalid or corrupted DOCX archive` , outputs a row with status `CORRUPTED_FILE` in the CSV, and continues processing remaining manuscripts.

Component C: Contributorship Section Discovery & Boundary Detection

Scenario C.1: Multi-Variant Section Heading Recognition

EDGE CASE

Intent: Detect contributorship blocks regardless of publisher-specific heading styles.

Validation: Correctly captures sections preceded by headings:

- `"Author Contributions"` (e.g., `merged_01.docx` , `merged_05.docx`)
- `"Author Contributions Statement"` (e.g., `merged_02.docx`)
- `"Contributors"` (e.g., `merged_03.docx`)
- `"CRediT Taxonomy"` (e.g., `merged_04.docx`)

Scenario C.2: False-Positive Rejection (Citations in References)

TRAP AVOIDANCE

Intent: Do not mistake bibliography entries or citations containing "Author Contributions" for the manuscript's contributorship section.

Validation: In `merged_13.docx` , where reference `[10]` contains *"Smith A. Author Contributions: a practical guide. J Pub Ethics. 2019..."*, the parser recognizes this is located inside the References block and does not falsely extract reference authors as contributors.

Scenario C.3: Handling Missing Contributorship Sections

NEGATIVE TEST

Intent: Accurately detect manuscripts that possess general acknowledgements or funding statements but omit author contributor statements.

Validation: In `merged_09.docx` and `merged_11.docx` , the tool logs `NO_CONTRIBUTION_STATEMENT_FOUND` rather than attempting to parse general lab acknowledgements as contributor roles.

Component D: Author Entity Resolution & Name Cleansing

Scenario D.1: Dotted Initials to Full Name Byline Mapping

EDGE CASE

Intent: Map abbreviated author initials in contributorship blocks to the full names extracted from the manuscript header byline.

Validation: In `merged_01.docx` , `J.D.` maps to *Jane Doe* and `A.B.` maps to *Anil Bose*. In `merged_08.docx` , `K.L.` maps to *Kai Lund*, `M.N.` to *Mira Nair*, and `P.Q.` to *Paul Quinn*.

Scenario D.2: Undotted Compact Initials Disambiguation

EDGE CASE

Intent: Map grouped, undotted multi-author initials to individual authors.

Validation: In `merged_07.docx`, `"AB and CD wrote and edited the paper. EF supervised the project."` resolves `AB` → *Anil Bose*, `CD` → *Carla Diaz*, and `EF` → *Erin Fox*.

Scenario D.3: Stripping Academic Titles, Affiliations & Footnote Noise

CLEANSING

Intent: Clean prefixes, numerical affiliations, and special symbols attached to author names.

Validation: In `merged_15.docx`, `"Dr. Jane Doe1*"` is sanitized to *Jane Doe* (stripping `Dr.`, affiliation `1`, and asterisk `*`), and `"Prof. Anil Bose2†"` is sanitized to *Anil Bose* (stripping `Prof.`, affiliation `2`, and dagger `†`).

Scenario D.4: Collective Author Distribution ("All Authors")

EXPANSION

Intent: Properly distribute contributor roles when statements use collective phrasing.

Validation: In `merged_17.docx` (*"All authors contributed equally to conceptualization, investigation and writing of the original draft"*), the tool extracts both header authors (*Jane Doe* and *Anil Bose*) and assigns each the stated roles.

Scenario D.5: Disjoint Multi-Line Author-Role Structure

LAYOUT

Intent: Associate authors and roles when formatted on alternating lines/paragraphs.

Validation: In `merged_08.docx`, where author initials (e.g. `K.L.`) sit on one line and roles (`Software, Validation`) sit on the line below, the parser maintains state and correctly binds roles to the preceding author.

Component E: Role Normalization & Semantic Mapping

Scenario E.1: Case-Insensitive & Delimiter-Agnostic Matching

STANDARD

Intent: Match canonical roles regardless of upper/lower case or delimiter style.

Validation: In `merged_10.docx`, `FORMAL ANALYSIS` maps to *Formal analysis*, `data curation` maps to *Data curation*, and semicolons/commas are normalized seamlessly.

Scenario E.2: Contribution Degree & Qualifier Stripping

EDGE CASE

Intent: Isolate the underlying CRediT role when qualified by contribution degrees (lead, equal, supporting).

Validation: In `merged_16.docx`:

- `"Conceptualization – lead"` → Role: *Conceptualization* | Degree: *lead*
- `"Methodology – equal"` → Role: *Methodology* | Degree: *equal*
- `"Writing – original draft – supporting"` → Role: *Writing – original draft* | Degree: *supporting*

Scenario E.3: Free-Form Narrative Semantic Mapping

NLP / SEMANTIC

Intent: Translate natural language descriptive verbs into official CRediT roles using live definitions.

Validation: Demonstrates high semantic fidelity across manuscripts:

- *"designed the study"* → **Conceptualization** (`merged_05.docx`, `merged_06.docx`)
- *"performed the experiments"* / *"collected the data"* → **Investigation** (`merged_05`, `merged_06`)
- *"analyzed the data"* / *"performed the statistical analysis"* → **Formal analysis** (`merged_05`, `merged_18`)
- *"wrote the manuscript"* / *"wrote initial draft"* → **Writing – original draft** (`merged_05`, `merged_06`)
- *"edited the final text"* / *"wrote and edited the paper"* → **Writing – review & editing** (`merged_07`, `merged_20`)
- *"acquired funding"* → **Funding acquisition** (`merged_19.docx`)
- *"administered the project"* → **Project administration** (`merged_19.docx`)
- *"provided resources"* → **Resources** (`merged_19.docx`)
- *"supervised the project / work"* → **Supervision** (`merged_07`, `merged_19`)
- *"Created the figures"* → **Visualization** (`merged_20.docx`)

Scenario E.4: Hybrid Narrative with Parenthetical Standard Roles

EDGE CASE

Intent: Prefer explicit standard roles enclosed in parentheses over ambient narrative text.

Validation: In `merged_20.docx`, *"Developed the analysis code (Software) and checked results (Validation)"* maps directly to **Software** and **Validation**.

Scenario E.5: Out-of-Taxonomy / Invalid Role Rejection

NEGATIVE TEST

Intent: Reject or flag actions that do not correspond to any valid scientific CRediT contributor role.

Validation: In `merged_18.docx`, for *"G.H. made the tea"*, the tool does NOT falsely hallucinate a CRediT role (e.g. Resources). It marks the contribution as `UNMAPPED / OUT_OF_SCOPE` with a low confidence flag.

Component F: Output Formatting & Auditability

Scenario F.1: Tabular CSV Schema & Multi-Role Decomposition

STANDARD

Intent: Produce clean, valid CSV output with full relational traceability.

Validation: The generated CSV includes: `Manuscript_ID`, `Author_Name`, `Author_Initials`, `Raw_Contribution_Text`, `Normalized_CRediT_Role`, `Mapping_Method` (Exact vs Semantic), and `Confidence_Score`. Authors with multiple roles generate multiple distinct rows or standardized semicolon-separated lists.

3. Master Manuscript Validation Matrix (Dataset: `merged_01.docx` – `merged_20.docx`)

The table below details each manuscript present in the evaluation suite, the specific pattern it contains, and the expected normalization outcome.

File	Byline Authors	In-Text Syntax / Pattern	Expected Mapped CRediT Roles
`merged_01.docx`	Jane Doe, Anil Bose	Initials with dots & commas: `J.D.: Conceptualization, Writing – original draft. A.B.: Methodology, Software.`	**Jane Doe:** Conceptualization; Writing – original draft **Anil Bose:** Methodology; Software

File	Byline Authors	In-Text Syntax / Pattern	Expected Mapped CRediT Roles
`merged_02.docx`	Jane Doe, Raj Patel	Heading variant (*Author Contributions Statement*); full names & semicolons.	**Jane Doe:** Conceptualization; Funding acquisition **Raj Patel:** Investigation; Data curation; Visualization
`merged_03.docx`	Maria Lopez, Tom Reid	Heading variant (*Contributors*); en-dash delimiter across paragraphs.	**Maria Lopez:** Supervision; Project administration; Resources **Tom Reid:** Validation; Writing – review & editing
`merged_04.docx`	Ken Ito, Lena Berg, Omar Nasser	Heading variant (*CRediT Taxonomy*); 3 authors listed across lines.	**Ken Ito:** Software; Validation **Lena Berg:** Formal analysis; Writing – original draft **Omar Nasser:** Resources
`merged_05.docx`	Jane Doe, Tom Reid	Free-form prose with full names: *"Jane Doe designed the study and analyzed the data. Tom Reid collected the data and wrote the initial draft."*	**Jane Doe:** Conceptualization; Formal analysis **Tom Reid:** Investigation; Writing – original draft
`merged_06.docx`	Jane Doe, Anil Bose	Free-form prose with initials: *"J.D. designed the study and wrote the manuscript. A.B. performed the experiments."*	**Jane Doe:** Conceptualization; Writing – original draft **Anil Bose:** Investigation
`merged_07.docx`	Anil Bose, Carla Diaz, Erin Fox	Grouped undotted initials: *"AB and CD wrote and edited the paper. EF supervised the project."*	**Anil Bose:** Writing – review & editing **Carla Diaz:** Writing – review & editing **Erin Fox:** Supervision
`merged_08.docx`	Kai Lund, Mira Nair, Paul Quinn	Alternating disjoint lines: Initial on line 1, roles on line 2 (`K.L.` → `Software, Validation`).	**Kai Lund:** Software; Validation **Mira Nair:** Visualization **Paul Quinn:** Methodology; Investigation
`merged_09.docx`	Sara Khan, Leo Moreau	Only generic acknowledgements present; NO contribution statement in manuscript.	*Status:* `NO_CONTRIBUTION_STATEMENT` (Zero hallucinated roles)
`merged_10.docx`	Rita Tran, Sam Ward	Case variations (`FORMAL ANALYSIS` , `data curation`) & only 1 author mentioned.	**Rita Tran:** Writing – original draft; Writing – review & editing; Formal analysis; Data curation **Sam Ward:** *Unlisted*
`merged_11.docx`	Mei Tanaka, Fatima Novak, Priya Khan	Standard acknowledgements only; NO contributorship statement.	*Status:* `NO_CONTRIBUTION_STATEMENT`
`merged_12.docx`	*N/A (Corrupted file)*	Corrupted binary document (not a valid zip/docx archive).	*Status:* `CORRUPTED_FILE_ERROR` (Graceful exception logging)
`merged_13.docx`	Lena Petrova, Hassan Novak, Vikram Ali	Trap case: Reference citation contains *"Author Contributions: a practical guide"*.	*Status:* `NO_CONTRIBUTION_STATEMENT` (Ignores citation trap)
`merged_14.docx`	Jane Doe, Anil Bose	Clean dotted initials with terminal periods: `J.D.: Investigation. A.B.: Supervision.`	**Jane Doe:** Investigation **Anil Bose:** Supervision
`merged_15.docx`	Jane Doe, Anil Bose	Academic titles & footnote noise: `Dr. Jane Doe1*: ... Prof. Anil Bose2†: ...`	**Jane Doe:** Conceptualization; Methodology **Anil Bose:** Supervision
`merged_16.docx`	Jane Doe, Anil Bose	Degree qualifiers: `A.B. (Conceptualization – lead; Methodology – equal)...`	**Anil Bose:** Conceptualization; Methodology **Jane Doe:** Writing – original draft

File	Byline Authors	In-Text Syntax / Pattern	Expected Mapped CRediT Roles
`merged_17.docx`	Jane Doe, Anil Bose	Collective attribution: *"All authors contributed equally to conceptualization, investigation..."*	**Jane Doe:** Conceptualization; Investigation; Writing – original draft **Anil Bose:** Conceptualization; Investigation; Writing – original draft
`merged_18.docx`	Erin Fox, Gus Hall	Out-of-taxonomy joke role: *"E.F. performed the statistical analysis. G.H. made the tea."*	**Erin Fox:** Formal analysis **G.H. (Gus Hall):** `UNMAPPED / REJECTED`
`merged_19.docx`	Mia Park, Sven Roth	Free-form administrative verbs: *"M.P. acquired funding... S.R. provided resources..."*	**Mia Park:** Funding acquisition; Project administration **Sven Roth:** Resources; Supervision
`merged_20.docx`	Hana Sato, Ivan Petrov, Julia Costa	Mixed: Verbatim roles, parenthetical roles `(Software)`, and narrative *"Created the figures"*.	**Hana Sato:** Conceptualization; Formal analysis; Writing – original draft **Ivan Petrov:** Software; Validation **Julia Costa:** Visualization; Writing – review & editing

4. Acceptance Criteria & Quality Metrics

- **Taxonomy Dynamic Compliance:** 100% of taxonomy roles must originate from the live runtime scrape of `credit.niso.org`. Static lists in code are rejected.
- **Author Cleansing Accuracy:** All honorifics (`Dr.`, `Prof.`), footnote markers (`1`, `2`, `*`, `†`), and trailing punctuation must be stripped before CSV output.
- **Initial Disambiguation:** 100% resolution rate for authors formatted as initials (`J.D.`, `AB`) against the byline authors.
- **Semantic Normalization Fidelity:** Common non-standard scientific phrases (e.g. *"wrote initial draft"*, *"performed experiments"*, *"analyzed data"*) must correctly map to their standard CRediT roles with high confidence (≥ 0.85).
- **Robustness & Non-Termination:** The parsing pipeline must process a directory of manuscripts in batch without stopping on single-file failures (e.g., corrupted archives like `merged_12.docx`).